MILITARY SPECIFICATION

TRAILER, MUNITION, AERO 51C

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of an 8500 pound capacity munitions trailer designated as AERO 51C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6866	Inspection, Penetrant Methods of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-G-10924	Grease, Automotive and Artillery
MIL-W-21157	Weldment, Steel, Carbon and Low Alloy (Yield Strength 30,000-60,000 Psi)
MIL-W-22248	Weldment, Aluminum and Aluminum Alloy
MIL-I-45208	Inspection System Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-278	Fabrication, Welding and Inspection, and Casting Inspection and Repair for
MIL-STD-281	Automobile, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of
MIL-STD-410	Non-Destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic, and Ultrasonic)
MIL-STD-831	Test Reports, Preparation of

2.1.2. Other Government documents, drawings, and Publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Naval Air Systems Command Data List 6SE00663, Trailer Munitions (and all documents listed thereon).

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Welding Society (AWS)
AWS D1.1 - Structural Welding Code

(Applications for copies should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 33125).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.1.1 Test report. A test report, delineating the results of first article test (see 4.3), shall be in accordance with MIL-STD-831.

3.2 Design and construction.

3.2.1 Construction. The trailer shall be manufactured in accordance with Data List DL6SE00663 and all documents listed thereon.

3.2.2 Welding. All welding shall be in accordance with MIL-W-22248 Class 3; MIL-STD-278, Class M; or MIL-W-21157, Class 5, as applicable.

3.2.2.l Welding inspection. During fabrication, all welding shall be inspected in accordance with MIL-I-6866 or MIL-I-6868 by personnel qualified in accordance with MIL-STD-410.

3.2.3 Lubrication, The trailer shall be lubricated with grease in accordance with MIL-G-10924.

3.3 Performance requirements.

3.3.1 Functioning. The trailer shall be examined for proper functioning of wheels, brake system and electrical system (see 4.6.2).

3.3.2 Overload. The trailer, when supported by its running gear, shall withstand a proof load of 17,000 (+100 -O) pounds, applied through the 2 sets of chocks positioned across the open hatchway doors (see 4.6.3).

3.3.3 Brake system.

3.3.3.1 Parking brakes. The parking brakes shall be capable of holding the trailer, loaded with a 8500 (+100 -0) pound test load, on a 10° grade in either direction (see 4.6.4.1).

3.3.3.2 Service brakes. The service brakes shall be capable of stopping the trailer, loaded with 8500 (+100 -0) pound test load, within 30 feet when traveling at 20 miles per hour, on level, dry concrete (see 4.6.4.2).

3.4 Towing force. The towing force required to move the trailer from rest on smooth, dry, level, paved surface, such as brushed concrete, shall not be greater than 50 pounds per ton of maximum gross weight, measured at the drawbar (see 4.6.5).

3.5 Mobility. The trailer shall be subjected to and pass the road test (see 4.6.6.1), the maneuverability test (see 4.6.6.2) and the side slope test (see 4.6.6.3) as specified herein.

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3.6 Workmanship. Workmanship shall be in conformance to the design and performance requirements specified herein. Equipment having missing, inoperative, defective, bent, broken or damaged parts, will not be acceptable. The installation of hardware parts, such as catches, pins, handles, nuts, bolts, etc., shall be accomplished in such a manner as to avoid-damaging the hardware or the mounting surface. Damaged hardware or mounting surfaces shall be touched up to provide a continuous protective coating. Screws, nuts, and bolts showing cross threading, mutilation or detrimental or hazardous burrs shall not be acceptable. Insulated wires shall be formed into cables or ducted whenever possible. Wires, cables, and brake lines shall be routed and positioned in a protective manner to avoid contact with rough or irregular surfaces and sharp edges. All welds shall be free of harmful defects such as cracks, porosity, undercuts, voids, and gaps. There shall be no evidence of burn through. Fillets shall be uniform and smooth. Angular or thickness misalignment, warpage, or dimensional change due to heat from the welding operation shall be within permitted tolerances. There shall be no damage to adjacent parts resulting from welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows :

a. First article inspection (see 4.3).
b. Individual inspection (see 4.4).
c. Sampling inspection (see 4.5).

4.3 First article inspection. The first article sample shall be subjected to the tests specified in Table I under first article and in the sequence as shown.

4.4 Quality conformance inspection. Each trailer shall be subjected to the tests specified in Table I under quality conformance, and in the sequence as shown,

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4.5 Sampling inspection. Trailer components shall be inspected using sampling plans as noted on the detail drawings or if none is specified, as directed by the procuring activity.

4.5.1 Rejection and retest. When a unit fails to satisfy the tests in Table I, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.5.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

TABLE I. First article and individual inspections.

Sequence	Test	Requirement	Test Paragraph	First Article	Quality Conformance
1	Examination of product	3.2-3.3.1	4.6.1	X	X
2	Functioning		4.6.2	X	X
3	Towing Force	3.4	4.6.5	X	
4	Breakaway		4.6.5.1	X	
5	Side Slope		4.6.6.3	X	
6	Service Brake	3.3.3.2	4.6.4.2	X	
7	Mobility	3.5	4.6.6	X	
8	Parking Brake	3.3.3.1	4.6.4.1	X	
9	Over Load	3.3.2	4.6.3	X	X

4.6 Methods of inspection.

4.6.1 Examination of product. Each trailer shall be examined visually to determine compliance with the requirements of this specification. Each trailer shall be inspected to determine conformance to the engineering drawings. Particular attention shall be given to routing protection and functioning of the lighting circuits and brake lines. Dimensions and attributes not in compliance with drawing tolerances shall not be acceptable.

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4.6.2 Functioning. Each trailer shall be examined to determine that wheels are correctly aligned, that all brake systems (including back-up) function and are correctly adjusted, that the electrical system is in working order, that all points requiring lubrication are properly lubricated, and that the wheel bearings are properly adjusted and lubricated. To check alignment, the trailer shall be on a level surface and with front wheels straight ahead. Measure the distance between the front wheels at hub height. Mark point where measurement was made. Roll trailer forward until these marks are on the opposite side of the axle (1800). Measure the distance between the wheels at hub height at this point. The difference between the two measurements is the actual toe-in; it should not be over 1/4 inch. If toe-in is not as stated, adjust to O to 1/4 inch. For adjustment, loosen jam nuts on both ends of the tie-rods and rotate tie-rods in or out as required. To adjust wheel bearings, jack or prop trailer to raise wheel off the ground. Spin the wheel and tighten adjusting nut until a heavy drag is felt. Then loosen nut until wheel rotates freely and bearing end play does not exceed 0.002 to 0.003 inches. Install cotter pin through nut and axle and replace grease cap.

4.6.3 Overload test. Place a 17,000 (+100 -0) pound test load across the open hatchway doors for not less than ten minutes. At completion of the test, there shall be no evidence of permanent distortion such as warpage of trailer deck, damage such as dents or breakage of components parts, misalignment of wheels, or other defects detrimental to the intended use.

4.6.4 Brake system tests. The trailer shall be loaded with a 8500 (+100 -0) pound test load consisting of bulk material (steel or concrete) placed across the open hatchway doors. The brake system shall then be tested as specified in 4.6.4.1 thru 4.6,4.2.2.

4.6.4.1 Parking brake tests. The trailer shall be loaded with a 8500 (+100 -0) pound test load and placed on a 10 degree inclined ramp, facing uphill. With parking brake applied, the trailer shall not roll. The loaded trailer shall then be tested facing downhill. This test should be performed after all mobility testing to insure that the brakes have been burnished sufficiently to achieve a successful test.

4.6.4.2 Service brakes. The trailer shall be loaded with a 8500 (+100 -0) pound test load and connected to a prime mover. The trailer test brakes shall then be tested as specified in 4.6.4.2.1 and 4.6.4.2.2.

4.6.4.2.1 Brake stopping capability test . Tow the loaded trailer at a speed of 20 MPH on a level, dry concrete surface. Stop by applying brakes of towing vehicle thus actuating the trailer inertia brake. The trailer shall stop within 30 feet or less of the point where brakes were first applied. The trailer shall not jackknife. At the end of five test cycles, inspect the service brakes and actuating mechanism. Failure to operate satisfactorily or evidence of excessive heating or distortion shall be cause for rejection.

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4.6.4.2.2 Back-up test. After passing the brake stopping capability test, the trailer shall be subjected to a back-up force. The service brakes shall hold momentarily, but with continued force, shall release and allow unrestricted rearward travel. The test shall be performed five times.

4.6.5 Towing force test. Place a scale between the prime mover and the towbar of the loaded trailer. The prime mover shall apply a gradual towing force through the scale to the loaded trailer, with its brakes released and with the trailer on level concrete. The trailer shall roll before the scale registers 270 pounds.

4.6.5.1 Breakaway test. Manually tow the loaded trailer and exert a pull on the breakaway cable to actuate the brake system. The wheels should lock, bringing the trailer to a halt.

4.6.6 Mobility tests. During the following tests, the trailer shall be closely observed for its ease of handling. Any undesirable tendency to yaw, sway, dog walk or jackknife shall be cause for rejection.

4.6.6.1 Road tests. The trailer, loaded with a 8500 (+100 - 0) pound test load, shall be subjected to the following road tests:

SURFACE	SPEED	DISTANCE
Level Paved Highway	20 - 25 MPH	100 Miles
Graded Gravel Road	10 - 12-1/2 MPH	50 Miles
*	10 - 12 MPH	5 Miles

*On a level course containing obstructions one inch high, regularly spaced at six foot intervals and at right angles to the direction of travel of the trailer.

4.6.6.2 Maneuverability. Upon completion of testing in 4.6.6.1, the loaded trailer shall be towed over level paved highway, at speeds of eight MPH through fifty right circle turns and fifty left circle turns, both at maximum cramping angle. It shall then perform twenty-five sudden stops at a speed of 20 MPH. At the conclusion of these tests and those described in 4.6.6.1, the vehicle shall be closely examined. Any evidence of excessive wear or damage shall be cause for rejection.

4.6.6.3 Side slope test. The trailer, loaded with a 8500 (+100 -0) pound test load, shall traverse at not less than an 8 degree (1-3/4 inch rise/ft) slope, at a speed of 5-10 MPH for a distance of 20 feet without overturning.

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4.7 Magnetic particle inspection test. All steel weldments and castings, when required by the detail drawing, shall be subjected to magnetic particle inspection described in MIL-I-6868. No cracks are permitted.

4.8 Penetrant inspection. All aluminum welded joints shall be inspected in accordance with MIL-I-6866. No cracks are permitted.

4.9 Non-destructive testing. Magnetic particle and dye penetrant testing as required, shall be performed on all stress weld areas at completion of all first article tests at the discretion of the Government inspector.

4.10 Inspection of packaging. The sampling and inspection of preservation and packing shall be in accordance with Section 5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-STD-281, level A or C, as specified.

5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The trailer, with applicable adapters, bolsters, cradles, and fixtures, is intended for use in transporting ordnance from storage to aircraft loading area. This utilization is intended to use to full advantage the capabilities of the AERO 47A loader.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaded property, if any, will be made available to the contractor.

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c. Selection of applicable levels of preservation and packaging required (see 5.1).

d. Whether first article inspection is required (see 3.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of documents first article.

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